Exercises: Linear Systems and Solution Theory – Answer Key

Mathematical Foundations for Machine Learning

Subspaces, Span, and Kernel

Exercise 1.1 [Easy]

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}$.

a) Compute $\ker(A)$.

b) Give a basis for $\operatorname{im}(A)$.

c) Verify the rank-nullity theorem for A.

Solution

a) We solve $Ax = 0$. Row-reducing A:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Back-substitution: $x_2 + x_3 = 0$ gives $x_2 = -x_3$, and $x_1 + 2x_2 + 3x_3 = 0$ gives $x_1 = -x_3$.

So $\ker(A) = \operatorname{span}\left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right\}$.

b) The pivot columns (1st and 2nd) of the original matrix give a basis for $\operatorname{im}(A)$:

$$\operatorname{im}(A) = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} \right\}.$$

c) We have $\operatorname{rank}(A) = 2$, $\dim(\ker(A)) = 1$, and $n = 3$. Indeed $2 + 1 = 3$. ✓

Exercise 1.2 [Medium]

Let $U, W \subseteq \mathbb{R}^n$ be subspaces.

a) Prove that $U \cap W$ is a subspace of $\mathbb{R}^n$.

b) Is $U \cup W$ necessarily a subspace? Either prove it or give a counterexample.

c) In a linear regression setting, suppose two groups of features $x_A \in \mathbb{R}^{d_1}$ and $x_B \in \mathbb{R}^{d_2}$ produce predictions via $y_A = Ax_A$ and $y_B = Bx_B$ respectively, where $A \in \mathbb{R}^{m \times d_1}$ and $B \in \mathbb{R}^{m \times d_2}$. Using part (a), explain why the set of predictions achievable by *both* feature groups is a subspace, and express it in terms of A and B.

Solution

a) We verify the three subspace axioms.

Contains zero: Since U and W are subspaces, $0 \in U$ and $0 \in W$, so $0 \in U \cap W$.

Closed under addition: Let $u, v \in U \cap W$. Then $u, v \in U$ and $u, v \in W$. Since U is a subspace, $u + v \in U$. Since W is a subspace, $u + v \in W$. Therefore $u + v \in U \cap W$.

Closed under scalar multiplication: Let $u \in U \cap W$ and $\alpha \in \mathbb{R}$. Then $\alpha u \in U$ (since U is a

subspace) and $\alpha u \in W$ (since W is a subspace). Therefore $\alpha u \in U \cap W$. $\square$
b) No. Counterexample: Let $U = \text{span}\{e_1\}$ and $W = \text{span}\{e_2\}$ in $\mathbb{R}^2$. Then $e_1 \in U \cup W$ and $e_2 \in U \cup W$, but $e_1 + e_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \notin U \cup W$, since it is neither a multiple of e_1 nor of e_2.
c) The set of predictions achievable by feature group A is $\text{im}(A)$, and similarly $\text{im}(B)$ for group B. The predictions achievable by both are exactly $\text{im}(A) \cap \text{im}(B)$, which is a subspace by part (a).

Exercise 1.3 [Hard]

Let $A \in \mathbb{R}^{m \times n}$. Prove that $\ker(A^T A) = \ker(A)$.

Hint: One inclusion is straightforward. For the other, consider $\|Ax\|^2$.

(ML context: This identity explains why the normal equations $A^T A\theta = A^T y$ have the same solution set structure as the original system $A\theta = y$ in least-squares regression.)

Solution

We show both inclusions.
($\subseteq$) $\ker(A) \subseteq \ker(A^T A)$**:** If $x \in \ker(A)$, then $Ax = 0$, so $A^T Ax = A^T(Ax) = A^T 0 = 0$. Thus $x \in \ker(A^T A)$.
($\supseteq$) $\ker(A^T A) \subseteq \ker(A)$**:** Let $x \in \ker(A^T A)$, i.e., $A^T Ax = 0$. We want to show $Ax = 0$. Compute:

$$\|Ax\|^2 = (Ax)^T(Ax) = x^T A^T Ax = x^T \cdot 0 = 0.$$

Since $\|Ax\|^2 = 0$ implies $Ax = 0$ (by positive definiteness of the norm), we get $x \in \ker(A)$.
Combining both inclusions: $\ker(A^T A) = \ker(A)$. $\square$
Consequence for ML: By the rank-nullity theorem, $\text{rank}(A^T A) = \text{rank}(A)$. This means $A^T A$ is invertible if and only if A has full column rank, which is the condition for the OLS solution $\hat{\theta} = (A^T A)^{-1} A^T y$ to be unique.

Rank, Image, and the Rank-Nullity Theorem

Exercise 2.1 [Easy]

A linear map $T : \mathbb{R}^7 \to \mathbb{R}^4$ has $\text{rank}(T) = 3$.

a) What is $\dim(\ker(T))$?

b) Is T surjective? Is T injective? Justify each answer.

c) How many free parameters does the general solution of $Tx = b$ have (when a solution exists)?

Solution

a) By the rank-nullity theorem: $\dim(\ker(T)) = \dim(\mathbb{R}^7) - \text{rank}(T) = 7 - 3 = 4$.
b) T is **not surjective**: surjectivity requires $\text{im}(T) = \mathbb{R}^4$, i.e., $\text{rank}(T) = 4$, but $\text{rank}(T) = 3 < 4$.
T is **not injective**: injectivity requires $\ker(T) = \{0\}$, but $\dim(\ker(T)) = 4 > 0$.
c) When a solution w exists, the general solution is $w + \ker(T)$. Since $\dim(\ker(T)) = 4$, the solution has **4 free parameters**.

Exercise 2.2 [Medium]

Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$.

a) Show that $\mathrm{im}(AB) \subseteq \mathrm{im}(A)$.

b) Conclude that $\mathrm{rank}(AB) \leq \min\{\mathrm{rank}(A), \mathrm{rank}(B)\}$.

Hint for the second inequality: use part (a) applied to $(AB)^T = B^T A^T$.

Solution

a) Let $y \in \mathrm{im}(AB)$. Then there exists $x \in \mathbb{R}^p$ such that $y = ABx$. Setting $z = Bx \in \mathbb{R}^n$, we have $y = Az$, so $y \in \mathrm{im}(A)$. This holds for all $y \in \mathrm{im}(AB)$, so $\mathrm{im}(AB) \subseteq \mathrm{im}(A)$.

b) From part (a): $\mathrm{im}(AB) \subseteq \mathrm{im}(A)$, so $\mathrm{rank}(AB) = \dim(\mathrm{im}(AB)) \leq \dim(\mathrm{im}(A)) = \mathrm{rank}(A)$.

For the other inequality, apply part (a) to $(AB)^T = B^T A^T$:

$$\mathrm{im}(B^T A^T) \subseteq \mathrm{im}(B^T).$$

Since $\mathrm{rank}(M) = \mathrm{rank}(M^T)$ for any matrix, we get:

$$\mathrm{rank}(AB) = \mathrm{rank}((AB)^T) = \mathrm{rank}(B^T A^T) \leq \mathrm{rank}(B^T) = \mathrm{rank}(B).$$

Combining: $\mathrm{rank}(AB) \leq \min\{\mathrm{rank}(A), \mathrm{rank}(B)\}$. $\square$

Exercise 2.3 [Hard]

Let $P \in \mathbb{R}^{n\times n}$ satisfy $P^2 = P$ (such a matrix is called a *projection*).

a) Show that $\mathrm{im}(P) = \ker(I - P)$ and $\ker(P) = \mathrm{im}(I - P)$.

b) Show that $\mathbb{R}^n = \mathrm{im}(P) \oplus \ker(P)$, i.e., every $x \in \mathbb{R}^n$ can be written uniquely as $x = u + v$ with $u \in \mathrm{im}(P)$ and $v \in \ker(P)$.

c) What are the possible eigenvalues of P? Use part (a) to justify.

d) Conclude that $\mathrm{rank}(P) = \mathrm{tr}(P)$.

(ML context: The hat matrix $H = X(X^T X)^{-1} X^T$ in linear regression is a projection. Its trace $\mathrm{tr}(H) = d$ equals the number of features, which measures model complexity.**)**

Solution

a) ($\mathrm{im}(P) \subseteq \ker(I - P)$): Let $y \in \mathrm{im}(P)$, so $y = Px$ for some x. Then $(I - P)y = y - Py = Px - P^2 x = Px - Px = 0$, so $y \in \ker(I - P)$.

($\ker(I - P) \subseteq \mathrm{im}(P)$): Let $y \in \ker(I - P)$. Then $(I - P)y = 0$, so $y = Py$. Thus $y \in \mathrm{im}(P)$.

The second identity follows by applying the first to $I - P$, which also satisfies $(I - P)^2 = I - 2P + P^2 = I - P$. So $\mathrm{im}(I - P) = \ker(P)$.

b) *Existence:* For any $x \in \mathbb{R}^n$, write $x = Px + (I - P)x$. We have $Px \in \mathrm{im}(P)$ and $(I - P)x \in \mathrm{im}(I - P) = \ker(P)$ by part (a).

Uniqueness: Suppose $x = u_1 + v_1 = u_2 + v_2$ with $u_i \in \mathrm{im}(P)$, $v_i \in \ker(P)$. Then $u_1 - u_2 = v_2 - v_1 \in \mathrm{im}(P) \cap \ker(P)$. If $w \in \mathrm{im}(P) \cap \ker(P)$, then $w = Pz$ for some z, and $Pw = 0$. So $0 = Pw = P^2 z = Pz = w$. Hence $u_1 = u_2$ and $v_1 = v_2$.

c) If $Px = \lambda x$ with $x \neq 0$, then $P^2 x = \lambda Px = \lambda^2 x$. But $P^2 = P$, so $\lambda x = \lambda^2 x$, giving $\lambda(\lambda - 1) = 0$. The only possible eigenvalues are $\lambda = 0$ and $\lambda = 1$. From part (a): the eigenspace for $\lambda = 1$ is $\ker(I - P) = \mathrm{im}(P)$, and for $\lambda = 0$ is $\ker(P)$.

d) Since the eigenvalues are only 0 and 1, we have $\mathrm{tr}(P) = \sum \lambda_i =$ (number of eigenvalues equal to 1) $= \dim(\mathrm{im}(P)) = \mathrm{rank}(P)$. $\square$

Solution Theory of Linear Systems

Exercise 3.1 [Easy]

Consider the system $Ax = b$ with

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}.$$

a) Form the augmented matrix $(A \mid b)$ and reduce it to row echelon form.

b) Does the system have a solution? If so, is it unique? Justify using the solution theory (i.e., check whether $b \in \operatorname{im}(A)$ and whether $\ker(A) = \{0\}$).

c) Write the full solution set in the form $w + \ker(A)$.

Solution

a) Augmented matrix and row reduction:

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 1 & 1 & 1 & 4 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 1 & -1 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

b) The system is consistent since $\operatorname{rank}(A) = \operatorname{rank}(A \mid b) = 2$, meaning $b \in \operatorname{im}(A)$. The solution is **not unique**: $\operatorname{rank}(A) = 2 < 3 = n$, so $\ker(A) \neq \{0\}$ (it has dimension $3 - 2 = 1$).

c) From the row echelon form, the free variable is $x_3 = t$:

$$x_2 = 1 + t, \quad x_1 = 3 - 2t.$$

One particular solution (set $t = 0$): $w = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$.

The kernel: $\ker(A) = \operatorname{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Full solution set:

$$\left\{ \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \;\middle|\; t \in \mathbb{R} \right\}.$$

Exercise 3.2 [Medium]

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Let $B \in \mathbb{R}^{m \times m}$ be an invertible matrix.

a) Prove that $Ax = b$ and $BAx = Bb$ have the same solution set.

b) Explain why this result justifies Gaussian elimination: every elementary row operation is a multiplication from the left by an invertible matrix, so the solution set is preserved.

c) Now let $C \in \mathbb{R}^{n \times n}$ be invertible. Does $Ax = b$ and $AC^{-1}y = b$ have the same solution set? If not, what is the precise relationship between the solutions?

Solution

a) Let $S_1 = \{x : Ax = b\}$ and $S_2 = \{x : BAx = Bb\}$.
($S_1 \subseteq S_2$): If $Ax = b$, then $BAx = Bb$, so $x \in S_2$.
($S_2 \subseteq S_1$): If $BAx = Bb$, multiply from the left by B^{-1}: $B^{-1}BAx = B^{-1}Bb$, which gives $Ax = b$, so $x \in S_1$.
Therefore $S_1 = S_2$. $\square$

b) Each elementary row operation (row swap, adding a scalar multiple of one row to another, scaling a row by $\lambda \neq 0$) corresponds to multiplying on the left by an invertible matrix (D_1, D_2, or D_3 respectively). Gaussian elimination is a sequence of such operations, i.e., it transforms $Ax = b$ into $(D^{(k)} \cdots D^{(1)})Ax = (D^{(k)} \cdots D^{(1)})b$. Since the product $B = D^{(k)} \cdots D^{(1)}$ is invertible (product of invertible matrices), part (a) guarantees the solution set is unchanged.
c) No, the solution sets are generally different. If x solves $Ax = b$, then setting $y = Cx$ gives $AC^{-1}y = AC^{-1}Cx = Ax = b$. So $y = Cx$ solves the second system. The relationship is: y solves $AC^{-1}y = b$ if and only if $x = C^{-1}y$ solves $Ax = b$. The solution sets are related by the invertible change of variables $y = Cx$, but they are **not the same set** (they live in the same space $\mathbb{R}^n$ but contain different vectors, unless $C = I$).

Exercise 3.3 [Medium]

Let $A \in \mathbb{R}^{m\times n}$. Consider the linear system $Ax = b$.

a) Show that the solution set of $Ax = b$ (when nonempty) is an affine subspace, i.e., it has the form $w + V$ where V is a subspace of $\mathbb{R}^n$.

b) Conversely, show that any affine subspace of $\mathbb{R}^n$ can be written as the solution set of some linear system.

Hint for (b): If V is a k-dimensional subspace, let $v_1, \ldots, v_k$ be a basis and construct a suitable matrix.

Solution

a) Let w be any particular solution, i.e., $Aw = b$. We claim the solution set equals $w+\ker(A)$.
($\subseteq$): If $Ax = b$, then $A(x - w) = Ax - Aw = b - b = 0$, so $x - w \in \ker(A)$, meaning $x \in w + \ker(A)$.
($\supseteq$): If $x = w + v$ with $v \in \ker(A)$, then $Ax = A(w + v) = Aw + Av = b + 0 = b$.
Since $\ker(A)$ is a subspace of $\mathbb{R}^n$ (it is always a subspace for any matrix), the solution set $w + \ker(A)$ is an affine subspace. $\square$
b) Let $S = w + V$ be an affine subspace, where V is k-dimensional with basis $\{v_1, \ldots, v_k\}$. We want to find A, b such that $\{x : Ax = b\} = S$.
Complete $\{v_1, \ldots, v_k\}$ to a basis $\{v_1, \ldots, v_k, u_1, \ldots, u_{n-k}\}$ of $\mathbb{R}^n$. Let $U = (u_1, \ldots, u_{n-k}) \in \mathbb{R}^{n\times(n-k)}$ and define the matrix $A = U^T \in \mathbb{R}^{(n-k)\times n}$ (so rows of A are $u_1^T, \ldots, u_{n-k}^T$). Set $b = Aw$.
Then $\ker(A) = V$ (because $Av_i = 0$ for all i and $\dim(\ker(A)) = n-\text{rank}(A) = n-(n-k) = k$), and the solution set of $Ax = b$ is $w + \ker(A) = w + V = S$.
Note: The rows of A need not be orthogonal to V; we only need $\ker(A) = V$. An orthogonal complement construction is one systematic way to ensure this.

Exercise 3.4 [Hard]

Consider a neural network layer performing the affine transformation $y = Wx + b$ where $W \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Suppose $\text{rank}(W) = r < \min\{m, n\}$.

a) Show that the set of all inputs $x \in \mathbb{R}^n$ that produce a given output $y_0 \in \text{im}(W) + b$ is an affine subspace of dimension $n - r$.

b) An adversarial perturbation δ is *undetectable by the layer* if $W(x+\delta)+b = Wx+b$. Show that the set of undetectable perturbations equals $\ker(W)$ and has dimension $n - r$.

c) A practitioner argues: "To reduce vulnerability to adversarial perturbations, we should use layers with $\text{rank}(W) = n$." Under what condition on m and n is this possible? What trade-off does this impose on the network architecture?

Solution

a) The equation $Wx + b = y_0$ is equivalent to $Wx = y_0 - b$. Setting $\tilde{b} = y_0 - b$, the condition $y_0 \in \mathrm{im}(W) + b$ means $\tilde{b} \in \mathrm{im}(W)$, so the system $Wx = \tilde{b}$ is consistent. Let w be any solution. By solution theory, the full solution set is $w + \ker(W)$, an affine subspace. Its dimension equals $\dim(\ker(W)) = n - \mathrm{rank}(W) = n - r$ by rank-nullity. $\square$
b) $W(x + \delta) + b = Wx + b$ simplifies to $W\delta = 0$, i.e., $\delta \in \ker(W)$. The set of undetectable perturbations is exactly $\ker(W)$, which has dimension $n - r$ by rank-nullity. $\square$
c) For $\mathrm{rank}(W) = n$, we need W to have full column rank, which requires $m \geq n$ (the output dimension must be at least as large as the input dimension). In this case $\ker(W) = \{0\}$, so there are no nonzero undetectable perturbations.
The trade-off: requiring $m \geq n$ at every layer means the network cannot **reduce dimensionality**, which is the purpose of many architectures (e.g., autoencoders, convolutional layers with fewer output channels). Full column rank throughout forces the hidden dimensions to be non-decreasing, increasing the parameter count and memory requirements.

Gaussian Elimination and LU Decomposition

Exercise 4.1 [Easy]

Bring the following matrix to row echelon form using Gaussian elimination. For each step, write down the corresponding elementary row transformation matrix.

$$A = \begin{pmatrix} 2 & 1 & -1 \\ -6 & -2 & 4 \\ 4 & 3 & -1 \end{pmatrix}$$

a) Perform the elimination and write A in row echelon form U.
b) Using the elementary matrices, write $A = LU$ where L is lower triangular.
c) What is $\det(A)$?

Solution

a) Step 1: Eliminate below the $(1, 1)$ pivot.

$R_2 \leftarrow R_2 + 3R_1$: Elementary matrix $D_2^{(1)} = I + 3e_2e_1^T = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

$R_3 \leftarrow R_3 - 2R_1$: Elementary matrix $D_2^{(2)} = I - 2e_3e_1^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$.

After Step 1: $\begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

Step 2: Eliminate below the $(2, 2)$ pivot.

$R_3 \leftarrow R_3 - R_2$: $D_2^{(3)} = I - e_3e_2^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$.

Row echelon form: $U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

b) We have $U = D_2^{(3)} D_2^{(2)} D_2^{(1)} A$, so:

$$L = (D_2^{(1)})^{-1}(D_2^{(2)})^{-1}(D_2^{(3)})^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}.$$

So $A = LU$ with $L = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

Verification: $LU = \begin{pmatrix} 2 & 1 & -1 \\ -6 & -2 & 4 \\ 4 & 3 & -1 \end{pmatrix} = A$. ✓

c) $\det(A) = \det(L) \cdot \det(U) = 1 \cdot (2 \cdot 1 \cdot 0) = 0$. Alternatively, $\det(A)$ is the product of the diagonal entries of U (since $\det(L) = 1$).

Exercise 4.2 [Medium]

Let $A \in \mathbb{R}^{n \times n}$ be invertible with LU decomposition $A = LU$.

a) Describe the algorithm for computing A^{-1} using the LU decomposition. How many forward and backward substitutions are needed?

b) What is the total arithmetic complexity of computing A^{-1} via LU, including the cost of the decomposition itself? Express your answer in terms of n.

c) Suppose you need to solve $Ax = b_i$ for k different right-hand sides $b_1, \ldots, b_k$. Compare the cost of:

(i) Computing A^{-1} and then multiplying $x_i = A^{-1} b_i$ for each i.

(ii) Reusing the LU decomposition to solve each system.

For what values of k (relative to n) does strategy (i) become more efficient?

Solution

a) To compute A^{-1} from $A = LU$, we need the n columns a_i^{-1} of A^{-1} satisfying $Aa_i^{-1} = e_i$. For each $i = 1, \ldots, n$:

- Forward substitution: solve $Lz_i = e_i$ for z_i (cost: $O(n^2)$)
- Backward substitution: solve $Ua_i^{-1} = z_i$ for a_i^{-1} (cost: $O(n^2)$)

Total: n forward and n backward substitutions.

b)

- LU decomposition: $\frac{2}{3}n^3$ operations.
- n forward substitutions, each $O(n^2)$: total $O(n^3)$.
- n backward substitutions, each $O(n^2)$: total $O(n^3)$.

Overall: $O(n^3)$ (dominated by $\frac{2}{3}n^3$ for the decomposition plus $\sim 2n^3$ for the substitutions).

c) Strategy (ii) with LU reuse: LU decomposition costs $\frac{2}{3}n^3$. Each solve costs $2n^2$ (one forward + one backward substitution). Total for k systems: $\frac{2}{3}n^3 + 2kn^2$.

Strategy (i): computing A^{-1} costs $\sim \frac{2}{3}n^3 + 2n^3 = \frac{8}{3}n^3$, plus k matrix-vector multiplications at $2n^2$ each. Total: $\frac{8}{3}n^3 + 2kn^2$.

Comparing: strategy (i) is cheaper when $\frac{8}{3}n^3 + 2kn^2 < \frac{2}{3}n^3 + 2kn^2$, which is never true (since $\frac{8}{3} > \frac{2}{3}$). Strategy (ii) is **always** at least as efficient, which is precisely why one should "never

solve a linear system using the inverse." Strategy (i) only makes sense when the inverse is needed for other purposes as well.

Exercise 4.3 [Medium]

Consider a matrix $A \in \mathbb{R}^{4\times 5}$ whose row echelon form is:

$$U = \begin{pmatrix} 3 & 1 & 0 & 2 & -1 \\ 0 & 0 & 5 & 1 & 3 \\ 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

a) What is $\operatorname{rank}(A)$?

b) Identify the pivot and free variables. What is $\dim(\ker(A))$?

c) Parameterize $\ker(A)$ by finding a basis (solve $Ux = 0$ using backward substitution with the free variables as parameters).

d) For which vectors $b \in \mathbb{R}^4$ does $Ax = b$ have a solution? Express the condition in terms of the components of b after applying the same row operations that reduce A to U.

Solution

a) $\operatorname{rank}(A) = 3$ (three nonzero leading coefficients, in columns 1, 3, and 5).

b) Pivot variables: x_1, x_3, x_5 (columns with leading coefficients). Free variables: x_2, x_4. $\dim(\ker(A)) = 5 - 3 = 2$.

c) Set $x_2 = s$, $x_4 = t$ (free parameters). Back-substitute in $Ux = 0$:

Row 3: $7x_5 = 0 \Rightarrow x_5 = 0$.

Row 2: $5x_3 + t + 0 = 0 \Rightarrow x_3 = -\frac{t}{5}$.

Row 1: $3x_1 + s + 0 + 2t + 0 = 0 \Rightarrow x_1 = -\frac{s+2t}{3}$.

So:

$$x = s\begin{pmatrix} -1/3 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -2/3 \\ 0 \\ -1/5 \\ 1 \\ 0 \end{pmatrix}.$$

A basis for $\ker(A)$ is $\left\{ \begin{pmatrix} -1/3 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2/3 \\ 0 \\ -1/5 \\ 1 \\ 0 \end{pmatrix} \right\}$ (or any scalar multiples to clear denominators).

d) Apply the same row operations to b, obtaining $\tilde{b} \in \mathbb{R}^4$. The system $Ux = \tilde{b}$ has a solution if and only if $\tilde{b}_4 = 0$ (the fourth component of the transformed right-hand side must be zero, since the fourth row of U is all zeros). Equivalently, $b \in \operatorname{im}(A)$ if and only if $\operatorname{rank}(A) = \operatorname{rank}(A \mid b)$.

Block Structure and Inverse via LU

Exercise 5.1 [Medium]

Compute the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$$

using the LU decomposition. Show all steps: find L and U, then solve $Lz_i = e_i$ and $Ua_i^{-1} = z_i$ for $i = 1, 2$.

Solution

LU Decomposition:
Eliminate below pivot: $R_2 \leftarrow R_2 - \frac{5}{2}R_1$.

$$U = \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ \frac{5}{2} & 1 \end{pmatrix}.$$

Column 1 of A^{-1}: Solve $Lz_1 = e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
Forward substitution: $z_{1,1} = 1$, then $\frac{5}{2}(1) + z_{1,2} = 0 \Rightarrow z_{1,2} = -\frac{5}{2}$.
So $z_1 = \begin{pmatrix} 1 \\ -5/2 \end{pmatrix}$.
Solve $Ua_1^{-1} = z_1$: backward substitution: $\frac{1}{2}a_{2,1}^{-1} = -\frac{5}{2} \Rightarrow a_{2,1}^{-1} = -5$. Then $2a_{1,1}^{-1} + (-5) = 1 \Rightarrow a_{1,1}^{-1} = 3$.
First column of A^{-1}: $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$.

Column 2 of A^{-1}: Solve $Lz_2 = e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
$z_{2,1} = 0$, $\frac{5}{2}(0) + z_{2,2} = 1 \Rightarrow z_{2,2} = 1$. So $z_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
Solve $Ua_2^{-1} = z_2$: $\frac{1}{2}a_{2,2}^{-1} = 1 \Rightarrow a_{2,2}^{-1} = 2$. Then $2a_{1,2}^{-1} + 2 = 0 \Rightarrow a_{1,2}^{-1} = -1$.
Second column of A^{-1}: $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$.
Result: $A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$.
Verification: $AA^{-1} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. ✓

Exercise 5.2 [Medium]

Consider the block-structured system:

$$\begin{pmatrix} D & B \\ C & A_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

where $D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 \end{pmatrix}$, $A_{22} = (2)$, $b_1 = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, and $b_2 = (4)$.

a) Compute the Schur complement $S = A_{22} - CD^{-1}B$.
b) Solve for x_2 using $Sx_2 = b_2 - CD^{-1}b_1$.
c) Back-substitute to find x_1.
d) Explain why exploiting the block structure is advantageous when D is a large diagonal matrix.

Solution

a) First compute $D^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}$.

$D^{-1}B = \begin{pmatrix} 1/2 \\ 1/3 \end{pmatrix}$.

$CD^{-1}B = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1/2 \\ 1/3 \end{pmatrix} = \frac{5}{6}$.

$S = 2 - \frac{5}{6} = \frac{7}{6}$.

b) Compute the modified right-hand side:

$D^{-1}b_1 = \begin{pmatrix} 3/2 \\ 2 \end{pmatrix}$, $CD^{-1}b_1 = \frac{3}{2} + 2 = \frac{7}{2}$.

$Sx_2 = b_2 - CD^{-1}b_1 = 4 - \frac{7}{2} = \frac{1}{2}$.

$x_2 = \frac{1/2}{7/6} = \frac{3}{7}$.

c) $Dx_1 = b_1 - Bx_2 = \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \frac{3}{7}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 18/7 \\ 39/7 \end{pmatrix}$.

$x_1 = D^{-1}\begin{pmatrix} 18/7 \\ 39/7 \end{pmatrix} = \begin{pmatrix} 9/7 \\ 13/7 \end{pmatrix}$.

Full solution: $x = \begin{pmatrix} 9/7 \\ 13/7 \\ 3/7 \end{pmatrix}$.

d) If $D \in \mathbb{R}^{n_1 \times n_1}$ is diagonal, then D^{-1} costs $O(n_1)$ and all products $D^{-1}B$, $CD^{-1}B$, $D^{-1}b_1$ cost $O(n_1 n_2)$. The Schur complement $S \in \mathbb{R}^{n_2 \times n_2}$ is small, so solving $Sx_2 = \tilde{b}$ costs $O(n_2^3)$. Total cost: $O(n_1 n_2 + n_2^3)$, compared to $O((n_1 + n_2)^3)$ for naive Gaussian elimination. When $n_1 \gg n_2$, the block method reduces cubic cost in n_1 to linear cost in n_1.

Mixed Challenging Exercises

Exercise 6.1 [Hard]

Let $A \in \mathbb{R}^{m \times n}$ with $m < n$ and $\operatorname{rank}(A) = m$ (full row rank).

a) Show that $Ax = b$ has a solution for every $b \in \mathbb{R}^m$.

b) Show that the solution is never unique.

c) Among all solutions, the *minimum-norm solution* is $x^* = A^T(AA^T)^{-1}b$. Show that $Ax^* = b$ and that $x^* \in \operatorname{im}(A^T)$.

d) Show that any solution $x = x^* + v$ with $v \in \ker(A)$, $v \neq 0$, satisfies $\|x\|^2 > \|x^*\|^2$.

Hint for (d): use the fact that $\operatorname{im}(A^T)$ *and* $\ker(A)$ *are orthogonal complements.*

Solution

a) Since $\operatorname{rank}(A) = m$, we have $\operatorname{im}(A) = \mathbb{R}^m$ (the image has full dimension). By solution theory, $Ax = b$ has a solution iff $b \in \operatorname{im}(A) = \mathbb{R}^m$, which holds for every b. $\square$

b) By rank-nullity: $\dim(\ker(A)) = n - \operatorname{rank}(A) = n - m > 0$ (since $m < n$). Since $\ker(A) \neq \{0\}$, the solution set $w + \ker(A)$ contains more than one element. $\square$

c) First, $AA^T \in \mathbb{R}^{m \times m}$ is invertible because $\operatorname{rank}(A) = m$ implies $\operatorname{rank}(AA^T) = m$ (using $\ker(A^TA) = \ker(A)$ applied to A^T, which has full column rank, so $\ker(A^T) = \{0\}$, giving $\ker(AA^T) = \{0\}$).

Then: $Ax^* = A \cdot A^T(AA^T)^{-1}b = (AA^T)(AA^T)^{-1}b = b$. ✓

Also, $x^* = A^T \underbrace{(AA^T)^{-1}b}_{\in \mathbb{R}^m} \in \operatorname{im}(A^T)$. ✓

d) Let $x = x^* + v$ with $v \in \ker(A)$, $v \neq 0$. Since $x^* \in \operatorname{im}(A^T)$ and $v \in \ker(A)$, and these are orthogonal complements (the fundamental theorem of linear algebra gives $\operatorname{im}(A^T) \perp \ker(A)$),

we have $\langle x^*, v\rangle = 0$.
Therefore:

$$\|x\|^2 = \|x^* + v\|^2 = \|x^*\|^2 + 2\langle x^*, v\rangle + \|v\|^2 = \|x^*\|^2 + \|v\|^2 > \|x^*\|^2$$

since $\|v\|^2 > 0$. This proves x^* has strictly smaller norm than any other solution. □

Exercise 6.2 [Hard]

In ridge regression, we solve $(X^TX + \lambda I)\theta = X^Ty$ for a regularization parameter $\lambda > 0$ and data matrix $X \in \mathbb{R}^{N\times d}$.

a) Show that $(X^TX + \lambda I)$ is always invertible for $\lambda > 0$, regardless of the rank of X.

b) What is $\ker(X^TX+\lambda I)$? What does this imply about the uniqueness of the ridge regression solution?

c) Show that the ridge solution can equivalently be obtained by solving the augmented system:

$$\begin{pmatrix} X \\ \sqrt{\lambda}\, I \end{pmatrix} \theta = \begin{pmatrix} y \\ 0 \end{pmatrix}$$

in the least-squares sense (i.e., minimizing $\|\tilde{X}\theta - \tilde{y}\|^2$ for the augmented matrix and vector).

d) What is the rank of the augmented matrix $\tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda} I \end{pmatrix} \in \mathbb{R}^{(N+d)\times d}$ when $\lambda > 0$? Use the solution theory to explain why the least-squares solution is unique.

Solution

a) Let $M = X^TX + \lambda I$. For any $v \neq 0$:

$$v^TMv = v^TX^TXv + \lambda v^Tv = \|Xv\|^2 + \lambda\|v\|^2 \geq \lambda\|v\|^2 > 0.$$

So M is positive definite, hence invertible. □

b) Since $M = X^TX + \lambda I$ is invertible, $\ker(M) = \{0\}$. The system $M\theta = X^Ty$ has exactly one solution: $\hat{\theta} = (X^TX + \lambda I)^{-1}X^Ty$. Uniqueness is guaranteed regardless of whether X has full column rank.

c) Define $\tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda} I \end{pmatrix}$ and $\tilde{y} = \begin{pmatrix} y \\ 0 \end{pmatrix}$. The least-squares objective is:

$$\|\tilde{X}\theta - \tilde{y}\|^2 = \|X\theta - y\|^2 + \lambda\|\theta\|^2$$

which is the ridge regression objective. Setting the gradient to zero:

$$\nabla_\theta = 2\tilde{X}^T(\tilde{X}\theta - \tilde{y}) = 2(X^TX + \lambda I)\theta - 2(X^Ty) = 0$$

giving $(X^TX + \lambda I)\theta = X^Ty$, the ridge normal equations. □

d) $\tilde{X} \in \mathbb{R}^{(N+d)\times d}$. The bottom block $\sqrt{\lambda} I \in \mathbb{R}^{d\times d}$ has rank d (since $\lambda > 0$). Therefore $\operatorname{rank}(\tilde{X}) = d$ (full column rank), which gives $\ker(\tilde{X}) = \{0\}$ by rank-nullity. By solution theory, the normal equations $\tilde{X}^T\tilde{X}\theta = \tilde{X}^T\tilde{y}$ have a unique solution (equivalently, $\tilde{X}^T\tilde{X} = X^TX + \lambda I$ is invertible, consistent with part (a)).

Exercise 6.3 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ and suppose Gaussian elimination *without* row swaps produces the LU decomposition $A = LU$.

a) Show that all leading principal minors of A are nonzero. (The k-th leading principal minor is $\det(A_{1:k,1:k})$.)

Hint: Use the block structure of L and U restricted to the first k rows and columns.

b) Conversely, suppose all leading principal minors of A are nonzero. Explain (without full proof) why this guarantees that Gaussian elimination can proceed without row swaps.

c) Give a 2×2 invertible matrix that requires a row swap (i.e., does not admit an LU decomposition without permutation). Verify that its first leading principal minor is zero.

Solution

a) Consider the LU decomposition $A = LU$ where L is unit lower triangular ($\ell_{ii} = 1$) and U is upper triangular with nonzero diagonal (since no row swaps were needed, all pivots are nonzero).
The first k rows and columns satisfy the block relation:

$$A_{1:k,1:k} = L_{1:k,1:k} \cdot U_{1:k,1:k}$$

where $L_{1:k,1:k}$ is unit lower triangular and $U_{1:k,1:k}$ is upper triangular with diagonal entries $u_{11}, \ldots, u_{kk}$ (all nonzero).
Therefore:

$$\det(A_{1:k,1:k}) = \det(L_{1:k,1:k}) \cdot \det(U_{1:k,1:k}) = 1 \cdot \prod_{i=1}^{k} u_{ii} \neq 0.$$

b) At step k of Gaussian elimination, the pivot element $a_{kk}^{(k)}$ (after eliminating the first $k-1$ columns) equals u_{kk} in the LU decomposition. The formula $\det(A_{1:k,1:k}) = \prod_{i=1}^{k} u_{ii}$ shows that $u_{kk} = \frac{\det(A_{1:k,1:k})}{\det(A_{1:k-1,1:k-1})}$. If all leading principal minors are nonzero, then every $u_{kk} \neq 0$, meaning no pivot is zero and no row swap is needed.

c) $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. This is invertible ($\det(A) = -1 \neq 0$). The first leading principal minor is $\det(A_{1:1,1:1}) = a_{11} = 0$. Gaussian elimination fails at the first step because the $(1,1)$ entry is zero, requiring a row swap.